

ARTICLE TYPE

Data-Driven Analysis of Musical Form and Harmonic Structure in AI-Generated Popular Music: A Case Study with Suno and Udio

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Abstract

We present a large-scale comparison of the formal and harmonic structures of music generated by users of two AI music generation platforms, Suno and Udio, against human-composed popular music. We address three questions: 1) How does AI-generated music’s harmony relate to human popular music? 2) What structural similarities or differences exist? 3) What considerations arise when applying music information retrieval (MIR) tools to AI-generated music? We analyze 20,002 Suno tracks, 20,002 Udio tracks, and 20,010 human-composed recordings. Harmonic analysis uses a state-of-the-art conformer-based chord estimation model; structural analysis uses a joint beat-tracking and semantic section labeling model (All-In-One). We model chord progressions as trigrams and apply byte-pair encoding (BPE) to section-label sequences to quantify formal diversity across the three collections. Analysis of over 2.52 million chord trigrams reveals that Suno concentrates its harmony on Western pop patterns such as I-V-vi-IV, while Udio exhibits more balanced chord distributions resembling human-composed music. BPE analysis shows Suno tends toward simple verse-chorus repetitions, whereas Udio exhibits the widest structural variety. These findings suggest that Suno (versions 3.5 and below) converges on a narrow harmonic and formal space, whereas Udio exhibits tendencies closer to human-composed music. We release beat-level chord annotations, metadata, section boundaries, self-similarity matrices, audio links, and code under an MIT license.

Keywords: AI Music, Generative AI, Musical Form, Harmonic Structure, Automatic Chord Estimation, Exploratory Data Analysis

1. Introduction

Commercial AI music generation platforms are becoming popular, and the resulting music is beginning to flood streaming services (Sturm et al., 2024). Suno,¹ launched online in late 2023, claimed in November 2024 to have a user base of 25 million people.² Udio,³ launched in the Spring of 2024, reported soon after that its users were generating ten songs a second.⁴ The summer of 2025 witnessed the appearance of an AI-generated music album by the fake “indie rock band” The Velvet Sundown, which has received millions of streams on Spotify.⁵ The French music streaming service Deezer announced in September 2025 that they are receiving 30,000 “fully AI-generated tracks daily”.⁶ This growing catalog of AI-generated music raises

several interesting questions for musicology and presents challenges for music information retrieval (MIR). Given the rate at which this catalog is growing, how reliable and scalable are the computational methods of MIR for addressing these questions? How does the harmonic content of AI-generated music relate to the harmonic practices found in human-composed music? What structural similarities and differences are there between AI-generated and human-composed music?

In this article, we address these questions specifically in the context of AI-generated music, drawing on examples from Suno and Udio. We describe the creation of a large corpus of 60,014 music recordings, balanced across three sources: 20,002 songs from Suno users, 20,002 songs from Udio users, and 20,010 human-composed Western popular songs.

Recent advances in MIR have made corpus-scale

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harmonic and structural analysis of popular music increasingly viable. Even though the challenges of resolving chord and structural information across diverse musical types remain an open problem, we employ consonance-ACE (Poltronieri et al., 2025) for harmonic analysis and All-In-One (Kim and Nam, 2023) for structural analysis, two recent implementations that bring the analysis significantly closer to musicological utility. The results of our harmonic analyses reveal that music from Suno exhibits strong stylistic characteristics of Western pop chord progressions, whereas music from Udio exhibits more balanced chord-quality distributions, resembling those found in human-composed music. The results of our structural analyses show that music from Suno tends toward simple AB (verse-chorus) repetition, whereas music from Udio exhibits a wider range of structural variations.

In summary, our contributions are: (i) a large comparative analysis of harmonic and formal properties in human-made and AI-generated music using a consistent pipeline, (ii) publicly released annotations for 60,014 tracks, including segmentations, beat-level chord estimates, style tags (Oramas et al., 2018), and audio metadata, and (iii) open-sourced code to reproduce the experiments.⁷

2. Background

The computational analysis of musical structure and harmony is a focus of MIR (Müller, 2015).

2.1 AI Music Generation

Recent open-source work on text-conditioned music and audio synthesis includes models such as MusicGen (Copet et al., 2023), MusicLM (Agostinelli et al., 2023), and Stable Audio Open (Evans et al., 2025), representing the state of the art in publicly available generative audio research. However, the focus of this paper is not on the generative models themselves, but on the musical properties of their output, analyzed at the corpus scale in comparison with human-produced music.

2.1.1 Evaluation of AI-Generated Music

Sturm and Ben-Tal (2017) evaluates folk-rnn, a generative system trained on a large corpus of folk tunes in ABC notation, examining outputs in relation to musical practice rather than only automatic metrics, with a concrete evaluation pipeline, corpus statistics analysis, and focused musical inspection to locate where model outputs resemble and where they diverge from human repertoire. Ariza (2009) shows that interrogator-style authorship discrimination tests only establish that a digital music artifact sounds plausible to listeners, which is insufficient to characterize the musical quality produced by an AI system.

Dervakos et al. (2021) propose a framework for evaluating symbolic AI-generated music using tools from music theory, defining heuristic metrics grounded in tonic coordinate systems and tone networks, such as

the circle of fifths, to quantify musicality without relying on user surveys or real data.

Wu and Yang (2020) analyze the outputs of the Jazz Transformer using objective metrics, including pitch-class histogram entropy, chord-progression irregularity, groove-pattern similarity, and structureness indicators, demonstrating that systematic shortcomings in AI-composed music are not fully captured by listening tests alone. Notably, their definition of chord progression irregularity as the proportion of unique chord trigrams in a piece establishes a precedent for trigram-based harmonic analysis of AI-generated music.

Some studies find biased perception of music when works are labeled as AI (Ragot et al., 2020; Shank et al., 2022; Déguernel and Sturm, 2023), others report null effects (Pasquier et al., 2016; Zlatkov et al., 2023). Together, these motivate a careful, data-driven comparison between music made with current AI platforms and music created by humans.

2.2 Music Structure Analysis

Music Structure Analysis (MSA) is the task of segmenting an audio recording into contiguous, non-overlapping sections and assigning labels that reflect musical function or similarity (Nieto et al., 2020). Computational approaches exploit four main principles: homogeneity (sections internally consistent with respect to some musical attribute), repetition (detecting recurrent sequences), novelty (identifying boundary points where musical content changes abruptly), and regularity (statistical properties of segment durations). Classical methods relied on self-similarity matrices and novelty curves derived from hand-crafted features such as chromagrams and MFCCs (Foote, 2000), repetition and recurrence-based representations (Levy and Sandler, 2008; Serrà et al., 2012), Laplacian spectral decomposition (McFee and Ellis, 2014), and barwise self-similarity with dynamic programming (Marmoret et al., 2023). Despite these advances, MSA remains an open challenge due to the subjectivity of structural annotations: the best reported structure labeling scores reach approximately 74% on the Harmonix Set (Kim and Nam, 2023), well below the estimated human inter-rater agreement of approximately 90% (Nieto et al., 2020).

More recent approaches have improved structural analysis through deep learning, including few-shot embeddings for Laplacian decomposition (Salamon et al., 2021), unsupervised contrastive multi-level representations (Buisson et al., 2022), self-supervised novelty detection (Peeters, 2023), hierarchical spectral-temporal Transformers for direct semantic labeling (Wang et al., 2022), and graph attention networks for pairwise structure prediction (Buisson et al., 2024).

For functional structure extraction, we employ the All-In-One model (Kim and Nam, 2023), which jointly performs beat tracking, downbeat detection, segment boundary detection, and semantic section labeling

(verse, chorus, bridge, etc.) in a single architecture. We selected it because it directly outputs the section labels our analysis requires and achieves state-of-the-art performance on the Harmonix Set across all four tasks.

2.3 Automatic Chord Estimation

Automatic Chord Estimation (ACE) is a central task in MIR with applications in music performance, transcription, and formal analysis research. ACE has evolved from early template and chroma-based systems with probabilistic sequence models, such as Hidden Markov Models, Conditional Random Fields, or Deep Belief Networks (Mauch and Dixon, 2009; Ni et al., 2012; Humphrey and Bello, 2015), to more recent deep learning approaches that learn end-to-end representations from spectral features, increasingly targeting large chord vocabularies that include sevenths, extensions, and inversions.

Large-vocabulary ACE has advanced through several complementary strategies. McFee and Bello (2017) decouple pitch-class detection from label assignment in a 170-class framework; Jiang et al. (2019) decompose chord labels into independent musical components (root, quality, bass, extensions), generalizing to 301+ classes; and Park et al. (2019) propose BTC, a bi-directional Transformer achieving competitive large-vocabulary performance with publicly available code. Most recently, Poltronieri et al. (2025) propose *consonance-ACE*, a Conformer-based model whose decomposed decoder reconstructs chord labels without a fixed vocabulary, paired with consonance-based label smoothing, outperforming BTC on both standard and non-binary evaluation metrics on RWC Pop and USPop. Waseem Akram et al. (2025) further extends this line with ChordFormer for large-vocabulary recognition.

Despite these advances, accessible implementations remain limited in coverage. Open-source libraries such as Madmom (Böck et al., 2016) and librosa (McFee et al., 2025) provide accessible implementations of baseline chord estimation algorithms, most of which remain fundamentally limited to detecting major and minor triads, struggling with tetrads, extensions, suspensions, and inversions. Earlier efforts to expand chord vocabularies, such as Chordino (Mauch and Dixon, 2010), while still functional as a Vamp plugin, suffer from platform- and dependency-compatibility issues; the library is no longer maintained and cannot be integrated into modern Python pipelines. Furthermore, analyses relying on chord-class aggregation can misrepresent stylistic progressions: Collins (2025) demonstrate that Chordino-based histograms failed to capture modal-interchange progressions while detecting only broad palette differences. Multi-source approaches such as DECIBEL (Odekerken et al., 2020), which fuse audio features with crowd-sourced MIDI annotations and guitar tablature, partially address vocabulary restrictions,

yielding improvements of 0.5–13.6 percentage points over audio-only baselines on 180 Beatles songs and 20 Queen songs from the Isophonics chord-annotation corpus⁸, though major and minor accuracy typically remains between 75% and 87%, a ceiling attributed to annotation subjectivity.

Roman Numeral Analysis (RNA) encodes chords relative to a tonal center, but benchmarks on the PARC dataset (Poppe et al., 2025) confirm that even state-of-the-art models achieve modest accuracy on real polyphonic audio, particularly for complex progressions.

Given these considerations, this study adopts *consonance-ACE* (Poltronieri et al., 2025) for three reasons: (1) its decomposed decoder is not constrained to a fixed vocabulary, accommodating harmonic diversity beyond major/minor triads; (2) consonance-based label smoothing produces errors that fall between harmonically related chords rather than random ones; and (3) its demonstrated superiority over BTC on standard benchmarks, combined with publicly available code, provides a reproducible foundation for large-scale analysis. The model is described in detail in Section 4.1.

2.4 Harmony Analysis of Corpora

Several annotated corpora and computational analyses have contributed to the study of harmony and musical structure in popular music. A hand-annotated corpus of rock songs provides detailed harmonic and melodic analysis, enabling statistical investigation of chords, scale degrees, and key-finding models (Temperley and Clercq, 2013). Expert ground-truth chord annotations for charting songs support research in automatic chord recognition and music analysis (Burgoyne et al., 2011). Computational analysis of Billboard chart data has been used to study the evolution of harmony, timbre, and style in popular music from 1960 to 2010 (Mauch et al., 2015). Long-term trends in harmonic and timbral features of contemporary Western popular music have also been examined to document changes in musical diversity (Serrà et al., 2012). Automated approaches to chord and key estimation provide frameworks for the simultaneous identification of chords and musical context from audio data (Mauch and Dixon, 2009). However, there is no comprehensive corpus-level comparison of harmonic and structural properties between current AI platforms and human music.

3. Corpora

The corpora are formed as a collection of three audio datasets (summarized in Table 1) containing audio files from Suno and Udio (AI-generated) and human-generated music from the Lastfm catalog. The audio tracks originate from (Cros Vila et al., 2025), where web scraping was performed for the Suno and Udio datasets between May and October 2024. Suno data was collected from its frequently updated “New Songs” playlist⁹, queried every two hours to capture its daily

influx of 50 new tracks. Udio data was collected using the platform’s search functionality, retrieving songs based on their popularity by tracking those with the highest number of plays in the 24 hours before each query (Cros Vila et al., 2025). Both datasets include metadata and audio files. The audio file folder contained more than 200,000 files, from which we randomly selected approximately 20,000 audio files from each dataset for analysis. The corpora uses the Million Song Dataset (MSD) (Bertin-Mahieux et al., 2011)¹⁰ to confirm the audio files are human-created music, leveraging Lastfm’s associated YouTube URLs to download around 200,000 unique tracks, where we randomly chose 20,010.¹¹ We thus refer to this collection as “Lastfm”.

Dataset	# JSON # Files	Duration Mean $\pm$ S.D.	Total Duration
Lastfm	20,010	261.3 $\pm$ 182.1s	1452h 6m 56s
Suno	20,002	175.2 $\pm$ 54.6s	1084h 47m 9s
Udio	20,002	143.2 $\pm$ 98.6s	801h 47m 36s

Table 1: The number of files is formed by a parallelized process with workers extracting form and chords data from a randomized folder of more than 100,000 audio files, with a limit of 20,000 JSON files

To address potential concerns about selection bias, we report user engagement metrics (likes and plays) across the Suno and Udio collections. Table 2 presents summary statistics, and Figure 1 displays their distributions on a logarithmic scale. Both platforms share identical median values (1 like and 4 plays), with means driven upward by a small number of high-engagement outliers. The 90th percentile reaches only 2 likes and 26 plays for Suno, and 3 likes and 32 plays for Udio, confirming that the vast majority of tracks in both collections received minimal engagement. A popularity-filtered dataset would yield a distinctive median difference relative to an unfiltered baseline; the equivalence observed here is consistent with unfiltered acquisition from each platform’s front page. No ranking, curation, or engagement threshold was applied to either collection. Although the Udio query nominally ranked tracks by 24-hour play count, the engagement distributions reported in Table 2 show that this signal operates at very low absolute magnitudes on Udio (median: 1 like, 4 plays). At these scales the ranking behaves as a near-uniform sample over recently uploaded content rather than as a popularity filter, which is consistent with the equivalence of central tendency between the two collections.

4. Method

This section describes the pipeline used to extract harmonic and structural information from the 60,014 audio recordings in our corpus, and the derived representations for comparative analysis.

Metric	Collection	Mean	Med.	P90	P95	Max
Likes	Suno	5.25	1	2	4	34,921
	Udio	1.83	1	3	8	898
Plays	Suno	204.88	4	26	50	445,490
	Udio	44.10	4	32	58	48,940

Table 2: Summary statistics of likes and plays for the Suno ($n = 19,972$) and Udio ($n = 19,992$) collections. P90 and P95 denote the 90th and 95th percentiles.

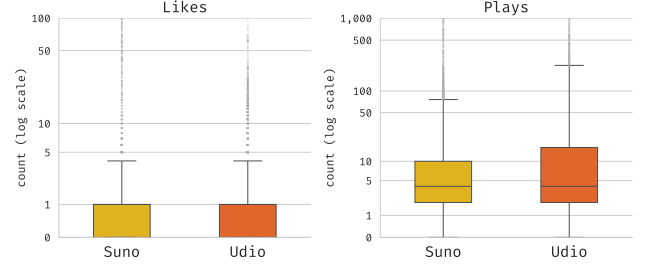


Figure 1: Box plots of likes and plays for the Suno and Udio collections on a logarithmic scale. Both platforms share a median of 1 like and 4 plays.

4.1 Automatic Chord Estimation

Chord estimation is performed using **consonance-ACE** (Poltronieri et al., 2025), which independently predicts root, bass, and pitch-class activations and reconstructs chord labels symbolically through a decomposed decoder. The model operates on CQT spectrograms computed over 20-second chunks and outputs frame-level chord predictions with no constraint to a fixed vocabulary. We use the publicly available pretrained model without modification. Chord segments shorter than 0.5 seconds are merged into adjacent segments, and consecutive frames with identical chord labels are collapsed into continuous intervals.

The output of the chord estimator consists of absolute chord labels (e.g., D:min7, G:7, C:maj7) aligned with a time onset-offset columns as shown in Table 3. To enable key-relative analysis, we convert these to functional harmony representations (e.g., ii, V, I). For each track, we first estimate the global key by computing the pitch-class distribution from the chord-root sequence and correlating it with the major- and minor-key profiles (Krumhansl and Kessler, 1982). Each absolute chord label is then mapped to its corresponding Roman numeral degree relative to the estimated key using the **music21** library (Cuthbert and Ariza, 2010). Consecutive identical Roman numerals are deduplicated before downstream n-gram analysis. The resulting chord intervals are subsequently aligned to beat positions estimated by the All-In-One model (Section 4.2), yielding the beat-level chord annotations included in the released dataset.

4.2 Musical Form Estimation

Musical structure is extracted using the All-In-One model (Kim and Nam, 2023), which jointly performs

Start (s)	End (s)	Chord
0.000	0.951	N
0.951	2.900	G:min
2.900	4.501	F:maj
4.501	7.471	D#:maj
7.471	9.118	G:min
9.118	11.346	F:maj
11.346	12.691	C:min
12.691	13.805	D#:maj
13.805	15.360	G:min
15.360	16.914	F:maj
16.914	18.840	D#:maj6
18.840	21.787	G:min
21.787	23.179	F:maj
23.179	24.617	D#:maj
24.617	25.499	G:min
25.499	26.241	D#:maj

Table 3: Example output of consonance-ACE. Each row contains the onset time, offset time, and estimated chord label. N denotes no chord detected.

four tasks in a single architecture: beat tracking (locating regular pulse positions), downbeat detection (identifying the first beat of each bar), segment boundary detection (finding time points where sections change), and functional structure labeling (assigning semantic labels such as verse, chorus, bridge, intro, outro, or solo to each segment).

The model first separates the input audio into four stems (drums, bass, vocals, other) using Hybrid Transformer Demucs (Rouard et al., 2023) and processes the resulting spectrograms through a stack of transformer modules combining dilated and neighborhood attention. Four output heads produce frame-level predictions for each task. The model achieves state-of-the-art performance on the Harmonix Set ($F1 = 0.958$ for beats, $F1 = 0.915$ for downbeats, $HR5F = 0.660$ for segmentation, $PWF = 0.738$ for structure labeling).

We apply the publicly available pretrained model without modification. For each track, the model outputs a sequence of labeled sections with their onset and offset times, stored as `_musical_form.lab` files. Consecutive identical section labels are subsequently merged before downstream analysis.

4.3 Trigram Dataset Generation

To systematically analyze harmonic tendencies across collections, we constructed a trigram dataset—a corpus of all consecutive three-chord sequences—extracted from each collection’s data. For example, the progression *Dm7, Ab7, Gm7, C#7, ...* yields *[Dm7, Ab7, Gm7]*, *[Ab7, Gm7, C#7]*, and so on. The process is as follows:

1. **Normalization:** Absolute chord labels are first converted to Roman numerals and consecutive identical Roman numerals are de-duplicated, as described in Section 4.1. The example above uses absolute chord labels for illustrative clarity. Sequences containing “no-chord” labels were not considered for trigrams.
2. **Trigram Generation:** For each song, we extracted

overlapping chord trigrams from the Roman numeral sequence using a sliding window approach:

$$(C_i, C_{i+1}, C_{i+2}), \quad i = 1, \dots, N-2$$

where N is the number of chord events in the song after de-duplication, and C_i denotes the i^{th} Roman numeral. All trigrams were retained, including repetitions and sequences spanning sections or boundaries, unless a song or section was shorter than three chords.

4.4 Byte-Pair Encoding of Harmonic and Formal Sequences

To compare structural and harmonic diversity across collections, we apply BPE (Sennrich et al., 2016) to two types of sequences. BPE is a sub-word tokenization algorithm that iteratively merges the most frequent adjacent symbol pairs in a corpus, building a vocabulary of recurring patterns from the bottom up.

For musical form, each track’s section sequence is first converted into overlapping trigrams of consecutive section labels (e.g. *verse · chorus · verse*), and BPE is applied using each trigram as the basic unit. BPE is run for up to 80 merge operations per collection, which is sufficient to exhaust all meaningful merges given the small section-label vocabulary.

For harmonic sequences, we first extract chord trigrams from the Roman numeral sequences produced by the ACE pipeline (Section 4.1). A trigram is a window of three consecutive, mutually distinct Roman numeral symbols. Each track’s sequence is represented as a series of trigrams, and BPE is applied at the trigram level: each token fed to the algorithm is itself a trigram, and merge operations identify the most frequent pairs of adjacent trigrams. BPE is run for 150 merge operations per collection, an empirical cap chosen to capture the most frequent recurring patterns without over-merging sparse ones. The resulting merge history, restricted to base-to-base merges (i.e., pairs in which both tokens are single trigrams), reveals the most common harmonic progressions of four chords and their cross-collection frequency distributions.

4.5 Axis of Awesome Chord Progression Analysis

We measure how often songs in each collection contain chord progressions matching the Axis of Awesome (Richards, 2017), using the Roman numeral sequences produced by the ACE pipeline. Four-chord sequences are formed as p-trigrams, pairs of adjacent chord trigrams, consistent with the trigram vocabulary used throughout this paper.

The Axis of Awesome is associated with three four-chord sets: the major form $\{I, V, vi, IV\}$, the minor form $\{i, bVI, III, bVII\}$, and a minor variant $\{i, bVI, bIII, bVII\}$. We treat all 24 orderings of each set as valid match targets, giving 72 distinct four-chord patterns in total. This decision is deliberate: the analysis targets harmonic vocabulary membership, whether a track

draws on axis-family pitch material, rather than the syntactic function of any specific ordering. As Richards (2017) discusses, rotations of the same four-chord set can carry different tonal implications; the reported axis rates should therefore be read as a measure of axis-family prevalence, not as evidence of a single recurring functional progression. From here, we will refer to this pattern of chord progressions as Axis.

All Roman numeral labels are first normalized enharmonically by consulting the estimated key and its accidentals to determine the correct spelling for each label (e.g., $\#V \rightarrow bVI$), and consecutive identical labels are removed. Trigrams are then extracted from each collapsed sequence using a sliding window of three chords; only trigrams in which all three chords are distinct are retained, consistent with the BPE trigram definition. Adjacent trigram pairs are formed from the retained list: a pair $(\langle a, b, c \rangle, \langle b, c, d \rangle)$ is valid only if the second trigram starts exactly one position after the first in the original sequence (i.e., they share a two-chord overlap). Each valid pair reconstructs a four-chord window (a, b, c, d) , which is tested for exact membership in the set of 72 axis patterns.

The p-trigram axis rate for a song is the proportion of valid p-trigram pairs whose reconstructed window matches any axis pattern, expressed as a percentage.

5. Comparing Patterns in Form and Harmony

The analyses proceed from the broadest view of the harmonic palette to increasingly specific patterns of chord succession and formal organization.

5.1 Trigram Tendencies Across Collections

Figure 2 shows the thirty most prevalent trigram patterns in the three members of the collection (Lastfm, Suno, and Udio), distributed over the ten most common styles found by extracting the metadata descriptors of the songs. The label “Others” recollects several styles in one group. The ‘Jazz’ annotation in Suno or Udio primarily applies to ambient instrumental music, with no particular harmonic patterns typical of standard jazz styles. Both Udio and Suno show a strong tendency to Pop, Rock, Metal, and Electronic.

While we acknowledge that the style tags associated with tracks in our dataset are inherently flawed and subject to user bias (users may arbitrarily assign genres, and the platforms’ own tagging systems lack rigorous musicological grounding), we use these classifications as they reflect the styles users most frequently specify when generating music. As demonstrated by (Casini et al., 2025), these same style descriptors (Pop, Rock, Electronic, Hip-hop, etc.) are among the most prevalent tags in the dataset.

The collection-level differences visible in Figure 2 are quantified in detail in Sections 5.2–5.4.

In Figure 3, an equivalent distribution of chord qualities is seen in all three collections. The major chord is the most common, followed by minor, power

chords, suspended fourths, and dominant sevenths. As an observation, dominant chords with sharp ninth or flat thirteenth, also altered dominant, are commonly used in minor and major modes, as a modal interchange in major tonalities. The dominant sharp eleventh is not common; the once-ubiquitous dominant substitution used in jazz, bossa-nova, or soul has become an endangered harmonic particularity in all three collections. Power chord, as a neutral quality, is used with mixed frequency across major and minor tonalities.

Figure 4 shows a distribution of the tonalities found in all collections. The quantification is sorted to Lastfm occurrence. The proportions show similarities on AI-generated music platforms; nevertheless, Suno shows stronger tendencies to produce C major, D major, G major, and B major, compared to Lastfm and Udio. Udio has a prominent presence of A minor, G minor and Eb minor. The distribution of tonalities is consistent with the cycle of the fifth.

Figure 5 reports the twenty most frequent chord trigrams found in the styles Rock, Electronic, Pop, Hip-Hop, Jazz (Chill-Out), Country, Blues, Soul, and Rhythm and Blues. Each collection has similarities and also strong tendencies revealing distinct harmonic signatures across Lastfm, Suno, and Udio. For example, in Soul, Udio and Suno are prone to harmonies based on dominant-subdominant permutations (V-I-IV, IV-V-I and I-IV-V).

5.2 Harmonic Tendencies

Figure 6 shows the relative frequency of canonical pop/rock chord trigrams — defined as permutations of $\{I/i, IV/iv, V/v, vi/VI/bVI\}$ — in each collection, computed as risk ratios against Lastfm and Udio using the method described in Section 4.3. The figure demonstrates that Suno tends to favor chord progressions strongly associated with the Western pop idiom, particularly those prevalent since the early 2000s. For every progression tested, Suno’s relative frequency is significantly greater than that observed in the other collections ($p < 0.001$ for all comparisons). As shown in (Temperley and Clercq, 2013), the centrality of progressions such as $I-V-vi-IV$, $IV-I-V$, and related idioms is a defining feature of Western pop/rock music; they report that the most frequent non-tonic chord roots are IV , V , $bVII$, and vi , with progressions among these chords forming the core stylistic harmonic vocabulary. Our results both replicate and extend these findings, revealing the extent to which such progressions are even more prevalent in AI-generated music from Suno.

To quantify the tendencies shown in Figure 5 for each canonical progression in Suno relative to the reference collections (Lastfm and Udio), we calculated a risk ratio (RR) for every trigram. Raw counts are misleading because Suno contains far more trigrams than the reference corpus, so the RR normalizes by each collection’s total:

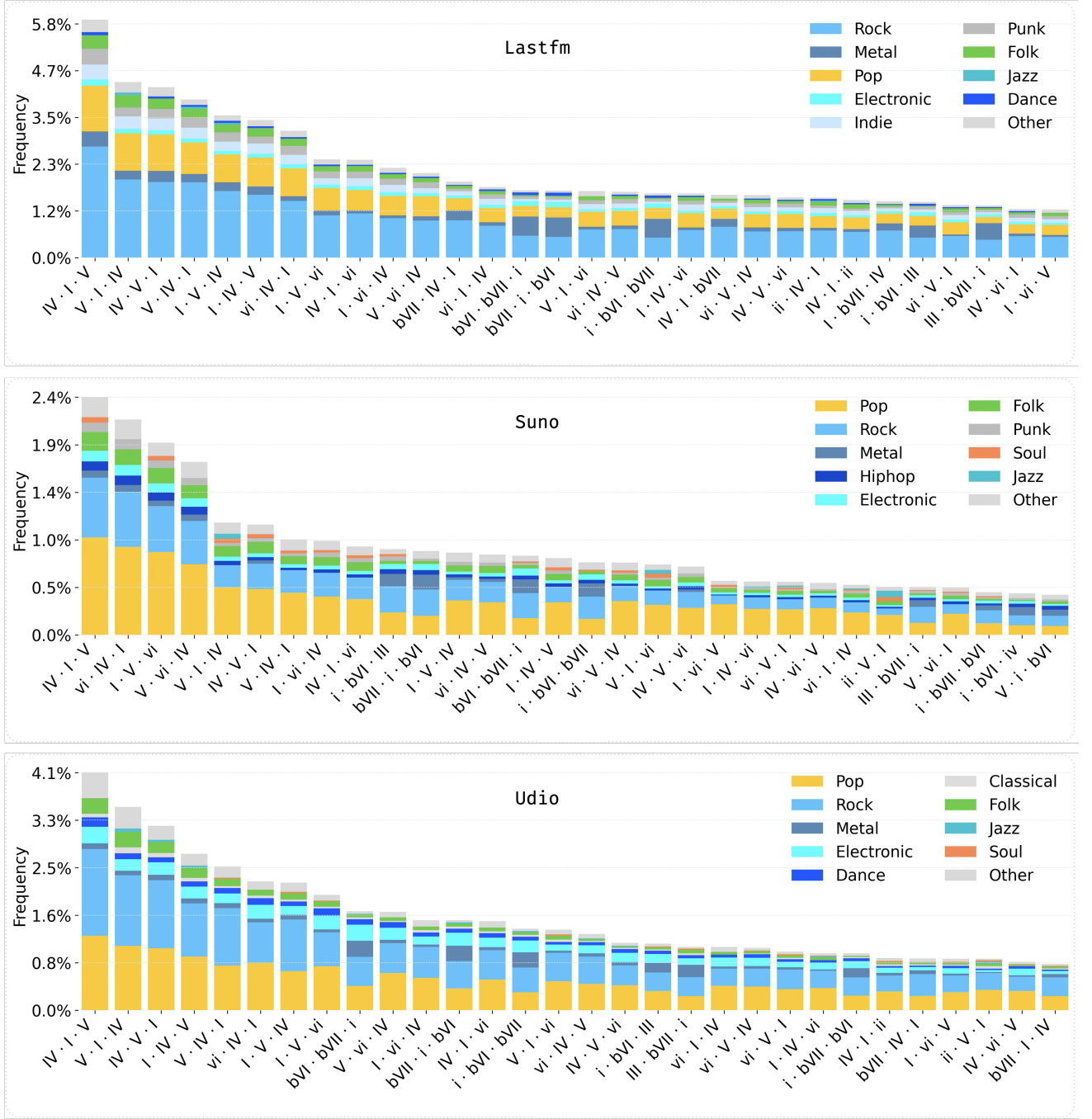


Figure 2: Top 30 most common trigrams found in Lastfm, Suno, and Udio datasets. The y-axis shows the relative frequency of each trigram in the collection. The distribution of styles differs across collections.

$$RR(t) = \frac{N_{\text{Suno},t}/N_{\text{Suno}}}{N_{\text{Ref},t}/N_{\text{Ref}}}$$

where $N_{\text{Suno},t}$ and $N_{\text{Ref},t}$ are the counts of trigram t in Suno and in the reference corpus, respectively, while N_{Suno} and N_{Ref} denote the total numbers of trigrams in those two corpora.

To test whether the difference is greater than expected by chance, we build a 2×2 contingency table (Suno vs. Reference $\times t$ vs. “all other trigrams”) and apply Fisher’s exact test. All raw p -values are cor-

rected for multiple comparisons with the Benjamini–Hochberg false-discovery-rate procedure (Benjamini and Hochberg, 1995); the resulting p -value is used to filter and choose the trigrams that are shown in the Table 4:

- at least 30 occurrences in Suno and 10 in the reference,
- risk ratio $RR \geq 2$,
- adjusted p -value < 0.01 .

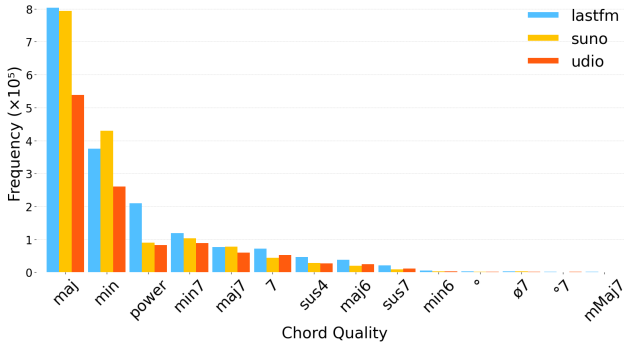


Figure 3: Vocabulary distribution of chord qualities over the collections. The y-axis indicates the total number of occurrences for each chord quality in each collection, scaled by $\times 10^5$.

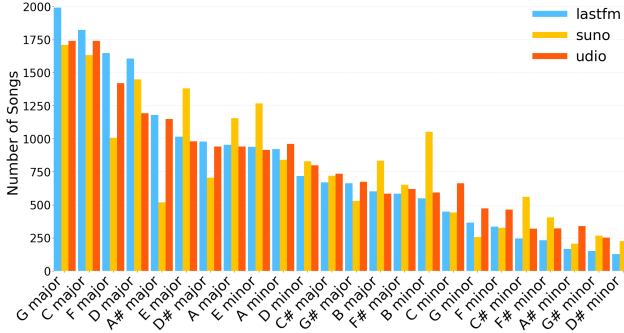


Figure 4: Distribution of most common tonalities in all three collections.

Trigram	C_{Suno}	C_{Udio}	C_{Last}	% Suno	% Udio	% Last	RR
Panel A: Top 15 canonical enriched trigrams							
vi, IV, I	34095	6442	8659	4.326	1.770	1.596	2.71
V, vi, IV	29383	6311	7311	3.728	1.734	1.347	2.77
I, V, vi	22803	3967	4948	2.893	1.090	0.912	3.17
V, i, bVI	4823	1221	1077	0.612	0.336	0.198	3.08
v, i, bVI	4610	863	1022	0.585	0.237	0.188	3.11
iv, V, i	4499	1373	1029	0.571	0.377	0.190	3.01
bVI, V, i	3508	1092	1074	0.445	0.300	0.198	2.25
iv, v, i	3287	653	739	0.417	0.179	0.136	3.06
V, i, iv	3273	1280	950	0.415	0.352	0.175	2.37
bVI, v, i	3117	618	658	0.396	0.170	0.121	3.26
i, v, bVI	2563	594	650	0.325	0.163	0.120	2.72
v, i, iv	2553	650	762	0.324	0.179	0.140	2.31
i, iv, V	2128	803	576	0.270	0.221	0.106	2.54
i, bVI, v	2081	554	555	0.264	0.152	0.102	2.58
i, iv, v	1929	581	622	0.245	0.160	0.115	2.14
Panel B: Top 10 non-canonical enriched trigrams (by RR)							
iv, ii, v	93	17	10	0.012	0.005	0.002	6.40
iii, vi, ii	1485	236	181	0.188	0.065	0.033	5.65
bVI, iv, ii	74	25	10	0.009	0.007	0.002	5.10
II, v, III	245	30	35	0.031	0.008	0.006	4.82
IV, iii, vi	1390	276	199	0.176	0.076	0.037	4.81
V, biv, vi	198	33	30	0.025	0.009	0.006	4.54
IV, biv, vi	131	29	21	0.017	0.008	0.004	4.30
biv, vi, IV	195	39	32	0.025	0.011	0.006	4.20
iii, IV, v	143	26	24	0.018	0.007	0.004	4.10
iii, ii, v	67	19	12	0.009	0.005	0.002	3.84

Table 4: Suno-enriched trigrams. Panel A lists the most prevalent canonical progressions, defined as any permutation of chords drawn exclusively from $\{I/i, IV/iv, V/v, vi/VI/bVI\}$. Panel B lists the highest RR non-canonical trigrams. C = raw occurrence count; RR = risk ratio (Suno / Lastfm). All rows are statistically significant (FDR-adjusted $p < 0.001$).

We applied BPE across the three music collections: 583
 859,882 trigrams in Lastfm, 1,052,255 trigrams in Suno, 584
 and 608,663 trigrams in Udio; in total, 2,520,800 tri- 585
 grams. This process finds and merges the most fre- 586
 quent adjacent tokens, revealing larger structural mo- 587
 tifs that reflect how songs are typically organized. Fig- 588
 ure 7 shows the most frequent BPE merge pairs; the 589
 results are discussed in Section 6.3. 590

5.3.1 Form Diversity and Style Differences 591

We also computed the “Unique Tokens Ratio,” which 592
 reflects how many different segment types appear in 593
 a song. Lower values suggest repetitive forms; higher 594
 values indicate more varied structures. This metric was 595
 lowest in Suno (0.44 ± 0.18) and highest in Udio ($0.58 \pm$ 596
 0.20), with Lastfm (0.51 ± 0.19) in between. 597

5.4 Axis of Awesome Tendencies 598

Table 5 reports the number and proportion of songs 599
 in each collection exceeding each p-trigram axis rate 600
 threshold. 601

At the $\geq 10\%$ threshold, 12.6% of Lastfm songs, 602
 14.7% of Udio songs, and 32.0% of Suno songs qual- 603
 ify as axis-based, confirming the findings of Richards 604
 (2017) on the cultural ubiquity of this progression and 605
 extending them to show that Suno reproduces it at a 606
 substantially higher rate relative to both the human 607
 reference corpus and Udio. 608

	Target trigram	Other trigrams
Suno	a	b
Reference	c	d

where $a = N_{\text{Suno},t}$, $b = N_{\text{Suno}} - a$, $c = N_{\text{Ref},t}$, and $d =$ 560
 $N_{\text{Ref}} - c$. A risk ratio $RR_t > 1$ together with $p < 0.01$ 561
 indicates that trigram t is significantly more common 562
 in Suno than in the reference corpus, as is the case for 563
 all the trigrams shown. 564

Table 4 shows a representative selection of the 118 565
 Suno-enriched trigrams passing all filters: Panel A lists 566
 the top 15 by Suno count of the 53 canonical progres- 567
 sions; Panel B lists the top 10 non-canonical trigrams 568
 by RR from the remaining 65. The columns provide the 569
 raw counts in Suno (C_{Suno}), Udio (C_{Udio}), and Lastfm 570
 (C_{Last}), their percentage of the total vocabulary, and 571
 the RR relative to Lastfm. 572

Across all canonical progressions in Panel A, RRs 573
 range from $2.1\times$ to $3.3\times$, with the major-mode axis 574
 chains (e.g., $I-V-vi$ at $3.2\times$, $vi-IV-I$ at $2.7\times$) among 575
 the most enriched. Every canonical pattern is signifi- 576
 cantly enriched ($p\text{-adj} < 10^{-16}$). 577

5.3 Byte-Pair Encoding Analysis of Segment Sequences 578

We apply BPE to sequences of label segments to ex- 579
 plore patterns in song structure. Each sequence begins 580
 with $\langle \text{init} \rangle$ and ends with $\langle \text{end} \rangle$ to mark the song’s 581
 boundaries. 582

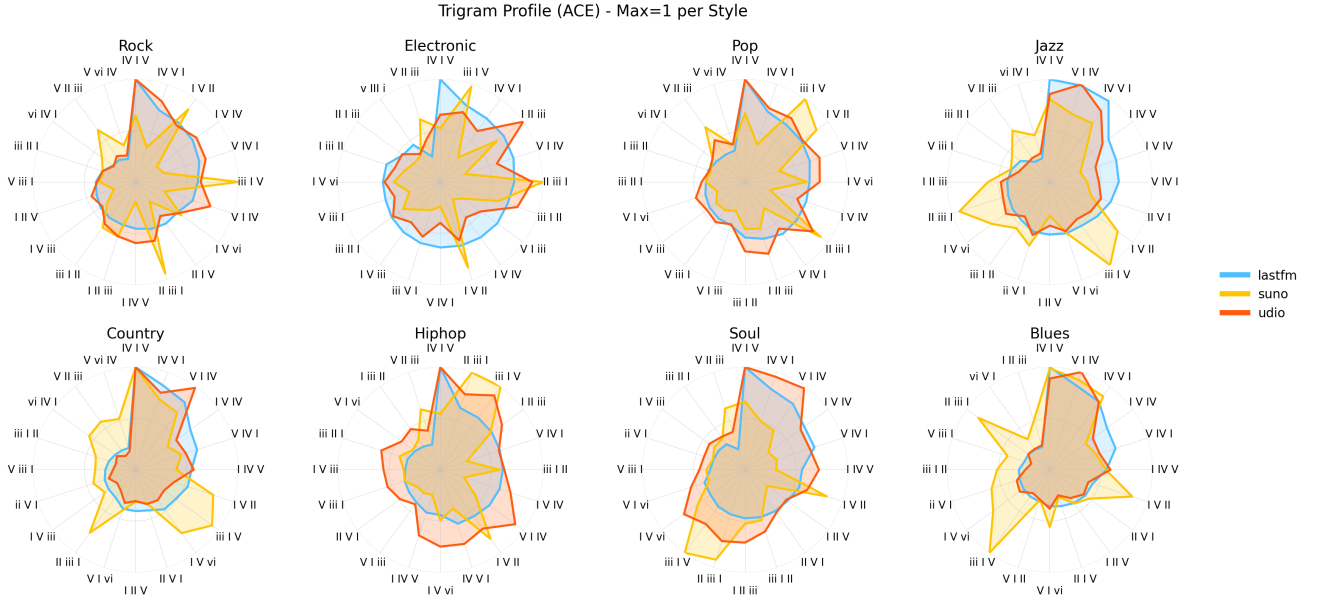


Figure 5: Spider-chart shows the common trigrams found per style in all collections. The spider chart is sorted by relevance, with Lastfm as the reference point.

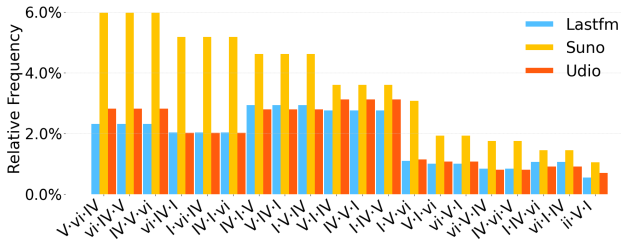


Figure 6: Relative frequency of canonical pop/rock trigrams in Suno compared with Lastfm and Udio. Bars show proportions of RRs, and all Suno-reference differences are significant at $p < 0.01$.

6. Discussion

We now discuss the three research questions about AI-generated music at the corpus scale and then draw out two cross-cutting implications: outcome homogenization and creative agency.

6.1 RQ1: Applying MIR Tools to AI-Generated Music

The chord estimator and structural model adopted in this study address the principal methodological concerns for corpus-scale analysis, but challenges remain that should be acknowledged. For chord estimation, even state-of-the-art models exhibit lower accuracy on complex chord types (sevenths, extensions, suspensions, inversions) than on triads (Pauwels and Peeters, 2013), and reported performance is sensitive to evaluation vocabulary and chord mapping strategies. Annotations from the Polyphonic Audio to Roman Corpus confirm that current RNA models achieve modest accuracy on real polyphonic audio, especially for harmonically complex progressions (Poppe et al., 2025). In our pipeline, the conversion from absolute chord labels to Roman numerals depends on reliable global key esti-

Table 5: Proportion of songs per collection at each p-trigram axis rate threshold. The axis rate is the fraction of a song’s four-chord windows that match any rotation of the Axis of Awesome progression. A song at $\geq 50\%$ has more than half its harmonic windows following the Axis loop; $\geq 10\%$ indicates the progression is a structural element of the song. Collections: Lastfm ($n = 18,916$), Suno ($n = 18,978$), Udio ($n = 19,031$).

Thresh.	Lastfm		Suno		Udio	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
>0%	5,034	26.6	8,864	46.7	5,066	26.6
$\geq 1\%$	4,950	26.2	8,809	46.4	5,023	26.4
$\geq 5\%$	3,355	17.7	7,225	38.1	3,719	19.5
$\geq 10\%$	2,392	12.6	6,076	32.0	2,791	14.7
$\geq 25\%$	1,195	6.3	3,987	21.0	1,431	7.5
$\geq 50\%$	434	2.3	1,993	10.5	562	3.0

mation via pitch-class correlation against Krumhansl-Kessler profiles. While this method is standard, key estimation errors propagate to every downstream Roman numeral label in a given track. For tracks with key ambiguity, modulation, or modal mixture, the functional analysis may misrepresent the intended harmonic relationships. Music21 library also required normalization of chord names to form the functional related chord. Furthermore, consonance-ACE assumes enharmonic equivalence: sharp ninths, sharp elevenths, and flat thirteenth are encoded as their enharmonic equivalents (flat third, flat fifth, and flat sixth respectively), and therefore do not appear as distinct labels in the chord vocabulary.

This study operates at the level of chord-label distributions and n-gram sequential patterns, which is the appropriate granularity for corpus-scale compar-

BPE Trigram Pair	verse-chorus-verse → chorus	9621	11320	6749
	chorus-inst-verse → chorus	2757	209	299
	verse-chorus-bridge → chorus	2272	1291	918
	verse-chorus-solo → chorus	1963	970	833
	verse-chorus-solo → verse	1738	424	772
	verse-chorus-inst → chorus	1166	1193	840
	verse-chorus-solo → outro	846	253	305
	verse-inst-verse → chorus	612	438	504
	intro-chorus-verse → chorus	594	0	277
	inst-chorus-inst → chorus	548	234	282
	intro-verse-chorus → outro	449	391	244
	verse-chorus-inst → solo	444	0	204
	chorus-bridge-verse → chorus	427	0	0
	verse-solo-verse → chorus	400	0	241
	verse-inst-verse → inst	0	0	182
	chorus-verse-chorus → outro	0	271	0
	intro-inst-solo → outro	0	238	0
	verse-chorus-inst → verse	0	2103	1599
	inst-chorus-verse → chorus	0	0	193
	verse-chorus-verse → inst	0	251	231
	verse-chorus-verse → solo	0	0	199
	intro-inst-verse → chorus	0	0	177
	verse-chorus-inst → outro	0	209	0
		Lastfm	Suno Collection	Udio

Figure 7: BPE merge-pair heatmap across collections.

Each row represents a frequently merged pair of consecutive musical-form trigrams (trigram A → last cell of trigram B, as the first two cells are redundant). Cell shading encodes relative frequency within each collection; integer annotations report absolute merge counts. Pairs are sorted by descending Lastfm frequency.

ison across 60,014 tracks. Voice leading, harmonic rhythm, and chord-voicing detail are not captured in any chord-symbol-based corpus study and fall outside the present scope.

For structural analysis, the All-In-One model provides semantic section labels (verse, chorus, bridge, intro, outro) unavailable from unsupervised methods. Its reported scores on the Harmonix Set ($HR.5F = 0.660$ for segmentation, $PWF = 0.738$ for structure labeling) remain below estimated human inter-rater agreement (Nieto et al., 2020), so a degree of error is present in the annotations. However, since the same model is applied uniformly across all three collections, these errors affect each corpus equally. Our analysis targets distributional tendencies across 60,014 tracks; systematic differences between collections therefore reflect genuine structural tendencies rather than tool-specific artifacts.

Despite these caveats, the combination of a chord estimator with a semantically aware structure analyzer represents a meaningful advance over the triad-level, label-free pipelines that were the practical default for large-scale audio analysis until recently. The rapid development of both ACE (Poltronieri et al., 2025; Park et al., 2019; McFee and Bello, 2017; Jiang et al., 2019; Waseem Akram et al., 2025) and MSA (Kim and Nam, 2023; Wang et al., 2022; Buisson et al., 2024; Peeters, 2023) tools, with publicly released code and pretrained models, makes corpus-scale harmonic and formal analysis of this kind increasingly feasible.

6.2 RQ2: Harmonic Relationship Between AI-Generated and Human-Composed Music

The four metrics used in this section capture different dimensions of harmonic practice and should be read accordingly. Chord-quality distributions (Figure 3) describe the harmonic palette, which chord types appear and how often. Trigram frequencies and cross-collection correlations describe local sequential behavior, in which three-chord successions recur and at what relative rates. Axis rates measure membership in a canonical four-chord family, not the prevalence of any single ordering. BPE merge histories describe recurring, longer patterns, across which tetragrams and extended sequences are consolidated within a collection. These dimensions are complementary but independent: a high trigram-frequency correlation between two collections, such as the $r = 0.952$ shared by Udio and Lastfm, indicates distributional similarity in local harmonic motion but says nothing about whether those progressions occupy the same formal positions within a song or occur at comparable rates of harmonic change.

Three strands of evidence converge on a consistent picture of how AI-generated harmony relates to human-composed music.

At the level of chord-quality vocabulary, the three collections are broadly similar: major triads dominate, followed by minor triads, power chords, seventh chords, dominant sevenths, and suspended chords (Figure 3). This indicates that current generative platforms reproduce the surface-level harmonic palette of popular music with fidelity. The differences emerge not in what chord types appear, but in how they are sequenced.

Trigram frequency analysis reveals that Suno concentrates its harmonic motion on a narrow subset of Western pop progressions. Across all 53 canonical progressions that pass the enrichment filters (FDR-adjusted $p \ll 0.001$, $RR \geq 1.3$), Suno is overrepresented at frequency ratios ranging from $1.3\times$ to $3.3\times$ relative to Lastfm. The highest-frequency 15 of these 53 progressions, listed in Panel A of Table 4, concentrate the overrepresentation at $2.1\times$ – $3.3\times$, while the remaining lower-frequency canonical progressions contribute ratios in the $1.3\times$ – $2.1\times$ range. The cumulative effect across all 53 enriched progressions is substantial: Suno’s harmonic vocabulary is contracted to 6,770 unique trigram types,

Musical Form Trigram Profile — Human-Annotated Styles (Normalized)

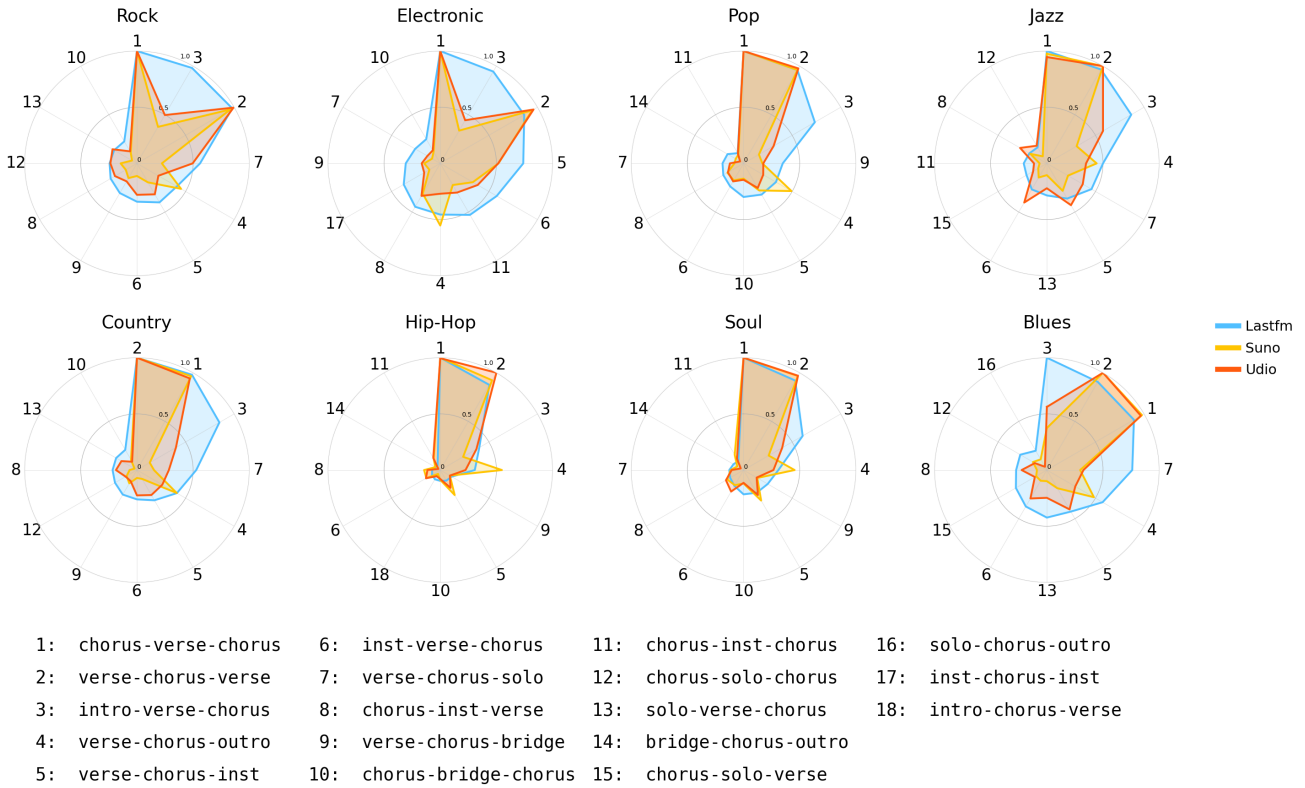


Figure 8: Normalized musical form trigram profiles across eight human-annotated genre tags per collection. Each axis represents a trigram of three consecutive section labels (e.g., chorus-verse-chorus), extracted after collapsing repeated adjacent sections. For each genre, the top trigrams are selected by Lastfm frequency, and values are normalized to the maximum count per collection within each genre.

compared to 8,056 in Udio and 9,838 in Lastfm.

The Axis of Awesome analysis extends this picture to the four-chord level. At the $\geq 10\%$ axis p-trigram threshold, 32.0% of Suno songs qualify as axis-based, versus 12.6% in Lastfm and 14.7% in Udio (Table 5). At the strictest level ($\geq 50\%$ of harmonic windows matching the Axis loop), Suno reaches 10.5% of songs, approximately 4.6 $\times$ the rate observed in Lastfm (2.3%) and 3.5 $\times$ that of Udio (3.0%). Songs built on canonical axis-family patterns are thus present at roughly 2.5 to 4.6 times the rate found in the human reference corpus, depending on the strictness of the threshold applied.

Udio’s harmonic distributions, by contrast, are statistically closer to those of Lastfm across all three measures. Cross-collection trigram frequency correlations confirm this: Udio and Lastfm share $r = 0.952$, while Suno correlates with Lastfm at only $r = 0.441$. The BPE merge histories of harmonic sequences (Figure 9) reinforce this at the tetragram level, with Udio and Lastfm sharing similar merge-pair frequency profiles while Suno diverges toward a smaller set of high-frequency patterns. Udio features minor-key functional progressions (e.g., **bVI-bVII-i**, **i-bVI-III**, **i-bVII-**

bVI) among its most frequent trigrams (see Figure 2). Progressions of this type, particularly the **bVI-bVII-i** pattern, have been identified as characteristic harmonic idioms of rock music, functioning as Aeolian cadences with subdominant–dominant–tonic implications distinct from conventional major-minor tonality (Biamonte, 2010). Lastfm’s top trigrams are dominated by functional major progressions (IV–I–V, IV–V–I, V–IV–I), with minor-mode patterns present but appearing further down the frequency ranking (see Figure 2).

6.3 RQ3: Structural Similarities and Differences

The BPE analysis of section-label sequences and the Unique Tokens Ratio provide complementary views of formal diversity across the three collections.

Suno exhibits the strongest convergence toward simple, repetitive structures. Verse-chorus cycles (ABAB patterns) dominate its merge history, and its Unique Tokens Ratio is the lowest of the three collections (0.44 ± 0.18), indicating that the average Suno track draws from fewer distinct section types than either human-composed or Udio-generated music. Udio, by contrast, shows the highest formal diversity (0.58 ± 0.20), with Lastfm occupying an intermediate

P-Trigrams	IV·I·V → vi -	2016	14341	1809	I·IV·V → I -	3015	3972	2577
	vi·IV·I → V -	2015	14233	1764	IV·V·I → IV -	2779	3825	2334
	I·V·vi → IV -	2031	14198	1745	V·I·IV → V -	2450	3490	2167
	V·vi·IV → I -	1855	13419	1671	I·V·IV → I -	3208	4236	2009
	I·vi·IV → I -	1898	4976	1253	IV·I·V → IV -	3113	4953	1982
	IV·I·V → IV -	3113	4953	1982	V·IV·I → V -	2902	4899	1880
	V·IV·I → V -	2902	4899	1880	IV·I·V → vi -	2016	14341	1809
	i·bVI·bVII → i -	1447	4614	1473	vi·IV·I → V -	2015	14233	1764
	bVI·bVII·i → bVI -	1361	4583	1307	I·V·vi → IV -	2031	14198	1745
	I·V·IV → I -	3208	4236	2009	V·vi·IV → I -	1855	13419	1671
	IV·I·vi → IV -	1628	4088	1088	i·bVI·bVII → i -	1447	4614	1473
	I·IV·V → I -	3015	3972	2577	V·I·IV → V -	1115	1763	1330
	vi·IV·V → I -	1073	3961	901	bVI·bVII·i → bVI -	1361	4583	1307
	IV·V·I → IV -	2779	3825	2334	IV·I·V → IV -	1123	1708	1261
	bVII·i·bVI → bVII -	1185	3757	1145	I·vi·IV → I -	1898	4976	1253
	vi·IV·I → vi -	1389	3668	913	bVII·i·bVI → bVII -	1185	3757	1145
	V·I·IV → V -	2450	3490	2167	IV·I·vi → IV -	1628	4088	1088
	V·I·vi → IV -	901	3254	839	vi·IV·I → vi -	1389	3668	913
	vi·V·IV → I -	965	3251	679	vi·IV·V → I -	1073	3961	901
	IV·V·I → vi -	899	3155	804	IV·V·vi → IV -	1114	2862	895
	i·bVI·III → bVII -	565	3150	631	V·I·vi → IV -	901	3254	839
	bVI·III·bVII → i -	550	3078	590	V·vi·IV → V -	1106	3075	829
	V·vi·IV → V -	1106	3075	829	IV·V·I → vi -	899	3155	804
	I·vi·IV → V -	870	2990	695	I·vi·IV → V -	870	2990	695
	vi·V·I → IV -	789	2923	603	I·IV·V → vi -	705	2480	682
	IV·V·vi → IV -	1114	2862	895	vi·V·IV → I -	965	3251	679
	III·bVII·i → bVI -	556	2839	618	I·bVII·IV → I -	1298	373	672
	bVII·i·bVI → III -	507	2792	520	I·vi·V → I -	743	1495	649
	I·IV·vi → V -	714	2696	449	i·III·bVII → i -	796	838	641
	V·I·IV → vi -	688	2656	479	i·bVI·III → bVII -	565	3150	631
	I·IV·V → vi -	705	2480	682	vi·IV·V → vi -	849	2088	629
	IV·vi·V → I -	690	2424	424	III·bVII·i → bVI -	556	2839	618
	I·V·vi → I -	872	2413	542	IV·vi·V → IV -	826	2110	613
	I·vi·V → IV -	782	2323	490	vi·V·I → IV -	789	2923	603
	IV·I·vi → V -	778	2309	591	IV·I·bVII → IV -	1030	292	592
	V·vi·I → IV -	673	2285	499	IV·I·vi → V -	778	2309	591
	V·IV·I → vi -	718	2233	526	bVI·III·bVII → i -	550	3078	590
	IV·vi·V → IV -	826	2110	613	i·#V·#VI → i -	491	1167	544
	vi·IV·V → vi -	849	2088	629	IV·vi·I → IV -	1049	1136	542
	vi·I·IV → V -	635	1931	512	I·V·vi → I -	872	2413	542
	lastfm	suno	udio		lastfm	suno	udio	

Figure 9: BPE treating the trigrams as cells. Two consecutive trigrams form a tetra-gram pattern called P-Trigram, where the last cell is shown as a simplification with the next consecutive function, taken from the second trigram (e.g. trigram [IV·I·V] paired with [I·V·vi]), becomes p-trigram [IV·I·V → vi]. The gradient of black and white shows a stronger prevalence in darker cells. The figure on the left is sorted using Suno as the reference. The right figure uses Udio as a reference.

position (0.51 ± 0.19). The BPE merge-pair heatmap (Figure 7) confirms that Udio’s merge frequencies are closer to Lastfm than to Suno. Of particular note, the second most frequent pattern in both AI collections — `verse·chorus·inst → verse` (Suno: 2,103; Udio: 1,599) — is entirely absent from Lastfm. Both systems frequently produce instrumental sections as structural connectors within verse-chorus cycles, a formal strategy with no parallel in the human corpus.

The normalized musical form trigram profiles per genre (Figure 8) reveal a consistent pattern across styles. Both Suno and Udio show a reduced range of formal options relative to Lastfm, with intro-verse-chorus sequences markedly under-represented and solo sections largely absent. This is consistent with how both platforms operate: users generate a song segment

that begins directly with a verse-chorus cycle and can later be extended via the web interface. The result is that intro sections, which precede the core structure, and solo sections, which typically appear later in human-composed music, are rarely generated in the initial output. The same genre-ontology caveat raised for Figure 5 applies here: the Jazz polygon in Figure 8 should not be read as a direct cross-collection comparison of jazz formal conventions, since the tracks grouped under that label differ substantially in character across the three corpora.

A key confound is song duration. Suno and Udio tracks average 175 and 143 seconds respectively, against 261 seconds for Lastfm (Figure 11). Shorter tracks have fewer opportunities for structural elaboration (bridges, instrumental breaks, key changes), so the reduced di-

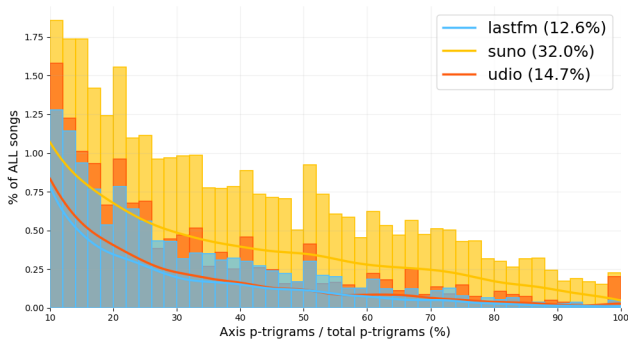


Figure 10: Distribution of the percentage of axis-based p-trigrams per song across the three collections (songs with $\geq 10\%$ axis content only). Lastfm shows 12.6%, Suno 32.0%, and Udio 14.7% of songs exceeding the 10% axis p-trigram threshold.

Suno exemplifies this pattern, converging on a statistical peak of canonical pop progressions (Figure 6, Table 4), also visible in the BPE of the most common p-trigrams (Figure 9). The output does not reflect the variety of harmonic practice or formal organization found in human-composed music. Several mechanisms may contribute to this convergence. First, training data bias amplification, in which generative models learn to reproduce only the most frequent patterns in their training corpora. Since contemporary pop music dominates streaming platforms and likely constitutes a significant portion of Suno’s training data, these harmonic patterns are overrepresented in its outputs. Second, optimization toward perceptual plausibility encourages models to generate music that sounds “familiar” to broad audiences, reinforcing existing harmonic conventions rather than exploring innovative progressions. Third, the lack of harmonic-prompting capabilities in current platforms means that users cannot directly specify harmonic content, instead relying on the models’ learned biases toward common progressions. Fourth, Suno’s developers may intentionally be targeting users who prefer a narrower spectrum of music, focusing on commercial modern pop styles. Our data cannot determine which of these factors is primary, but the contrast with Udio is informative. Udio produces more diverse harmonic and structural outputs than would be expected from a presumably comparable training distribution, suggesting that model architecture and generation strategy play a role beyond the training data alone. These platforms promise to make music creation widely accessible, but generating convincing audio recordings does not necessarily entail preserving musical knowledge or variety. We note that a restricted harmonic vocabulary is not in itself a compositional deficiency: twelve-bar blues, punk, and minimal techno are all built on narrow chord sets by deliberate stylistic choice. Our claim is different: the data show that Suno applies a single harmonic template across all stated genres, a platform-level convergence that is independent of user intent. This is consistent with Suno’s own positioning: the platform’s leadership has described it explicitly as gamifying musical creation.¹² This raises questions about whether these tools genuinely support creative agency among musicians with any degree of musical knowledge (Wiggins, 2007, 2016; Lin et al., 2023; Tan, 2024). The prevalence of simple, repetitive formal structures (as revealed by our BPE analysis shown in Figure 9) further suggests that current platforms may be optimized to generate familiar, commercially viable content rather than to support experimentation or skill acquisition. This optimization strategy may serve commercial interests, but it limits the creative potential of AI as a compositional tool.

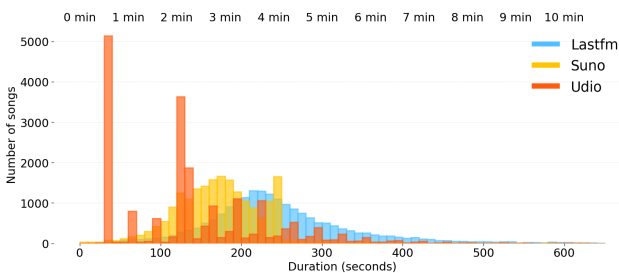


Figure 11: Distribution of song durations across the three collections (Lastfm, Suno, Udio). The x-axis represents duration in seconds, and the y-axis indicates the number of songs per 10-second bin.

A further caveat concerns formal labels and musical function. The All-In-One model assigns labels based on acoustic properties, but a verse-chorus alternation in a generated track may represent static loop repetition rather than the narrative-arc form those labels imply in conventional songwriting (Nieto et al., 2020). Distinguishing conventional pop form from non-teleological loop structures would require additional features such as melodic contour, rhythmic analysis, or lyric structure.

6.4 Outcome Homogenization and Creative Agency

The harmonic compression documented in RQ2 and the structural narrowing documented in RQ3 both point toward a form of outcome homogenization, consistent with the concept of algorithmic monoculture, where shared components across AI systems increase the likelihood of convergent outputs (Bommasani et al., 2022).

7. Conclusion

Across all three research questions, a consistent picture emerges. Applied uniformly to 60,014 tracks, current MIR tools reveal that the two AI platforms under study diverge from each other as substantially as they diverge from human-composed music. Suno concentrates its harmonic output on a narrow subset of Western pop progressions, overrepresented at 2.5 to 4.6 times the rate found in the human reference corpus, and its formal repertoire is dominated by simple verse-chorus cycles with little structural variety. Udio, by contrast, exhibits harmonic and structural distributions that are statistically closer to those of human-composed music across every measure examined: trigram frequency correlations, axis-family rates, BPE merge profiles, and formal diversity all converge on the same ordering, Udio closer to Lastfm, Suno further away. Because both platforms draw on popular music at a corpus-level scale, this divergence implicates platform-level choices, model architecture, generation strategy, and optimization objectives as primary determinants of the harmonic and formal character of AI-generated music. This narrowness emerges despite diverse user intent. The analysis from Casini et al. (2025) shows that Suno and Udio users actively prompt with distinct style descriptions, alongside explicit decade- and era-conditioning, indicating that the contracted harmonic vocabulary documented here is a property of the model's generation rather than of homogeneous user demand or restricted training exposure.

The most pressing methodological extension is to improve functional harmonic labeling to handle modulation, modal mixture, and key ambiguity, thereby sharpening analyses currently dependent on a single global key estimate per track. Tracking how these patterns evolve across successive model versions would further test whether the divergence documented here is a stable property of each platform or a moving target.

More broadly, this work demonstrates that fully automated, open-source MIR pipelines can now support corpus-scale harmonic and formal analysis at a scale and vocabulary richness that were not previously practical. The combination of rich-vocabulary chord estimation and semantic section labeling, applied here to 60,014 tracks, offers a replicable methodology for computational musicology and MIR as AI-generated music continues to grow in both volume and diversity.

8. Reproducibility

We release the complete dataset (60,014 JSON files with beat-level chord annotations, segmentation boundaries, style metadata, and audio links) and all analysis code under an MIT license.¹³

9. Ethics and consent

Although the full ramifications of AI-generated material remain uncertain, the sheer speed and volume at which these systems can create new works demand

scholarly attention (Sturm et al., 2024; Pelly, 2025). Because our study centers on music, we gathered examples from the commercial services Suno and Udio. Doing so raises a set of ethical issues. Even when a site's terms of use do not appear to permit automated collection, the legitimacy of repurposing that material for research—particularly under a "fair use" rationale—deserves scrutiny. As a publicly funded team investigating AI music and its broader consequences, we rely on the text-and-data-mining exception laid out in the EU Copyright Directive 2019/790 (European Union, 2019) to justify our data acquisition.

Notes

- ¹ <https://suno.com>, last accessed Sep. 30 2025.
- ² <http://bit.ly/4nnZA55>, last accessed Sep. 30 2025.
- ³ <https://www.udio.com>, last accessed Sep. 30 2025.
- ⁴ <https://bit.ly/46Q0Gyy>, last accessed Sep. 30 2025.
- ⁵ <https://www.theguardian.com/technology/2025/jul/14/an-ai-generated-band-got-1m-plays-on-spotify-now-music-insiders-say-listeners-should-be-warned>, last accessed Sep. 30 2025.
- ⁶ <https://www.musicbusinessworldwide.com/nearly-a-third-of-all-tracks-uploaded-to-deezer-are-now-fully-ai-generated-says-platform/>, last accessed Sep. 30 2025.
- ⁷ https://anonymous.4open.science/r/Aimir_Structure_Analysis
- ⁸ <http://isophonics.net/content/reference-annotations>, last accessed Sep. 30 2025.
- ⁹ <https://suno.com/playlist/cc14084a-2622-4c4b-8258-1f6b4b4f54b3>, last accessed Sep. 30, 2025.
- ¹⁰ https://github.com/renesemela/lastfm-dataset-2020/raw/master/datasets/lastfm_dataset_2020/lastfm_dataset_2020.db, last accessed Sep. 30 2025.
- ¹¹ We used yt-dlp <https://github.com/yt-dlp/yt-dlp>, last accessed Sep. 30 2025.
- ¹² Adam Neely: Suno, 'AI Music, and the Bad Future'. <https://www.youtube.com/watch?v=U8dcFhF0Dlk>. Last accessed May 05, 2026
- ¹³ https://anonymous.4open.science/r/Aimir_Structure_Analysis/README.md

Acknowledgements

Contributions

XXX did ... YYY did ...

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